

Chirag S Vadrali

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PROFESSIONAL SUMMARY

ECE student (CGPA: 8.15) with industry experience in PCB design, signal integrity, and embedded systems. Proficient in KiCad — schematic capture, 4-layer impedance-controlled stackups, DFM, and Gerber generation. Hands-on with SI/PI principles: differential pair routing, return-path analysis, crosstalk mitigation, decoupling networks. Seeking a hardware validation/SI internship to deepen expertise in high-speed digital interfaces and PCB characterization.

EXPERIENCE

Hardware Design Intern | VortexFlow Labs 2026 – Present

- Designed 4-layer impedance-controlled stackups (signal–GND–PWR–signal) in KiCad; applied SI/PI best practices — return-path continuity, decoupling placement, and crosstalk isolation across high-speed traces.
- Routed controlled-impedance traces (50 Ω) for GNSS modules; validated signal integrity via bench testing and simulation.
- Executed DFM reviews, design rule checks, and Gerber generation for manufacturer handoff; collaborated cross-functionally to document and communicate design constraints.
- Developed embedded firmware in C/Python for I2C/UART sensor integration on ESP32/ESP8266; wrote data-acquisition scripts directly applicable to test automation workflows.

PROJECTS

GPS Module - Reverse Engineering – Schematic Recovery · GNSS · Continuity Tracing

- Reverse-engineered a commercial u-blox-class GNSS module to a full schematic/netlist using continuity tracing and datasheet cross-referencing of RF front-end and power-supply sections..

4-Layer ESP32 IoT Development Board – Kicad Designer · SI/PI · EMC · High-Speed Design

- Engineered 4-layer PCB with signal–GND–PWR–signal stackup for EMC compliance and power integrity; applied return-current path analysis, via stitching, and decoupling networks to minimise crosstalk and ground bounce on USB/SPI differential pairs.

13-Cell Li-ion Battery Management System – TI BQ769x2 · Cell Balancing · Schematic-to-layout

- Designed a 13S BMS end-to-end (schematic through layout) using the TI BQ769x2 analog front-end, implementing passive cell balancing and OV/UV/OC/short-circuit protection.
- Performed power-net layout with current-path sizing and decoupling for the AFE, balancing the high-voltage stack against noise sensitive sensing line.

IoT Environmental Monitoring Board (4-Layer) – ESP8266 · KiCad · Power Integrity · Multi-Sensor Fusion

- Designed a 4-layer PCB integrating temperature/humidity, pressure, and air-quality sensors on shared I2C/SPI buses with isolated analog sensing nodes.
- Applied decoupling networks and ground-plane partitioning to isolate sensitive analog sensor readings from digital switching noise.

TECHNICAL SKILLS

PCB Design	KiCad, Altium Designer (Basic) — schematic capture, layout, DFM, Gerber generation, 4-layer impedance-controlled stackups
SI / PI	50 Ω impedance design, differential pair routing, return-path analysis, decoupling networks, EMC, crosstalk mitigation
High-Speed I/F	SPI, I2C, UART, USB peripheral routing, BLE HID; familiarity with PCIe Protocols; NavIC/GNSS interface integration
Embedded / FW	C, C++, Python — BLE HID stack, UART/I2C/SPI drivers, sensor fusion; ESP32/ESP8266
Power	Li-ion BMS (13S), LDO regulation, Buck regulation, decoupling design, onboard charging circuits
Tools	Linux, Git, KiCad DRC, Altium DRC

EDUCATION

B.E. Electronics & Communication Engineering – Sep 2023 – May 2027

BLDEAs College of Engineering & Technology, Bijapur – **CGPA: 8.12 / 10**

Coursework: Digital Electronics, Electronic Circuit Analysis, Electromagnetic Theory, Signals & Systems, Microprocessors & Embedded Systems, PCB Manufacturing Processes, IoT, Computer Architecture